

The impact of soil carbon management on soil macropore structure: a comparison of two apple orchard systems in New Zealand

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Summary

We analysed the long-term effect of the addition of organic carbon (C) on the macropore structure of topsoils. For this purpose we compared the top 50 mm in the tree rows of an organic apple orchard with those in an adjacent conventional orchard with the same soil type, texture and previous land-use history in New Zealand. After 12 years the topsoils of the organic orchard had 32% more soil organic carbon (SOC) sequestered than those of the conventional, integrated orchard because of regular compost applications and grass coverage. We quantified the macropore structure (macropores = pores > 0.3 mm) of nine undisturbed soil columns (43 mm long, 20 × 17 mm in the plane) within each orchard using 3D X-ray computed tomography. The macroporosity ($7.5 \pm 2.1\%$) of the organic orchard soil was significantly greater than that of the integrated orchard ($2.4 \pm 0.5\%$). The mean macropore radius was similar in the organic and integrated systems, with 0.41 ± 0.02 mm and 0.39 ± 0.01 mm, respectively. The connectivity of macropores tended to be greater in the organic than in the integrated system, but this was not statistically significant. The pronounced soil C management in the organic orchard increased both the formation of macropores by roots and a larger fresh weight of anecic earthworms, and the stabilization of the macropore structure was increased by a larger aggregate stability and microbial biomass compared with those of the integrated orchard. We simulated the diffusion through the measured pore structures of segments of the soil columns. The segments had the length of the mean aggregate size of the soils. The relative diffusion coefficients at this aggregate scale were significantly greater in the organic (0.024 ± 0.0009) than in the integrated (0.0056 ± 0.008) orchard. In a regression analysis with both the porosity and connectivity of macropores as significant variables, 76% of the variability of the relative diffusion coefficients was explained in the integrated, and, with the porosity as the only significant factor, 71% of the variability in the organic orchard. We hypothesize that a greater relative diffusion coefficient at the aggregate scale would reduce nitrous oxide (N₂O) production and emission in a wet soil and suggest that soil C management combats climate change directly by sequestering C and indirectly in the form of a reduction of N₂O emissions, by creating more macropores.

Introduction

Different definitions exist for the size of macropores, to distinguish them from other pores. Following others (Jarvis, 2007), we defined macropores as pores with a diameter larger than 0.3 mm. The formation and stabilization of macropores are driven by processes that are linked to the soil's carbon (C) management. Macropore networks are formed by the combined

activities of roots and soil fauna such as earthworms (Bastardie *et al.*, 2003) and by the swelling and shrinking of the soil as a result of drying and wetting cycles. The stability of macropore networks depends on the stability of the aggregates, which is mainly a function of fine roots and microbial activities (Le Bissonais, 1996; Le Bissonais & Arrouays, 1997).

It is well known that soil organic carbon (SOC) is a key property of soil quality (Glover *et al.*, 2000; Debosz *et al.*, 2002; Sparling & Schipper, 2004; Fliessbach *et al.*, 2007). Soil quality is the capacity of a specific kind of soil to function, within

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natural or managed ecosystem boundaries, to sustain plant and animal productivity, maintain or enhance water and air quality, and support human health and habitation (Karlen *et al.*, 1997).

Additionally, with climate change, it has been suggested the sequestration of C has become an important ecosystem service of soils (Lal, 2004; Smith, 2004). Several strategies exist to prevent the loss of SOC, to enhance soil C sequestration, and by this to conserve or increase soil quality. For example, long-term additions of organic C (OC) such as the regular application of compost and manure to soils, or the use of a grass cover on a bare soil, are known to increase SOC contents (Foley & Cooperband, 2002; Celik *et al.*, 2004; Deurer *et al.*, 2008). Generally, all 'land management practices that maintain or increase SOC' have been defined as 'soil carbon management' (Kimble *et al.*, 2007).

Previous studies have analysed the effect of long-term additions of OC and of the increase of SOC on soil structure. Soil structure describes the arrangement of the solid parts of the soil and of the entire pore space located between them (Marshall & Holmes, 1979). In most of those studies, 'structure' was defined to be equivalent to the soil's aggregate structure, neglecting the soil's macropores outside aggregates. For example, additions of OC and/or the increase of SOC led to a larger mean weighted diameter of aggregates and/or to a better aggregate stability (Gupta & Germida, 1988; Carter, 1992; Le Bissonais & Arrouays, 1997; Franzluebbers, 2002; Hernandez-Hernandez & Lopez-Hernandez, 2002; Celik *et al.*, 2004; Chung *et al.*, 2008). However, it is increasingly being recognized that such quasi-stability measures form only a small part of the important properties of soil. The undisturbed 3D macropore structure is, however, a key property of the soil's structure. Yet, little is known about how it is modified by long-term additions of OC.

Macropores are important for several soil processes and functions. For example, macropores can trigger preferential flow, the fast direct transfer of contaminants such as herbicides from the soil surface to the groundwater (Jarvis, 2007). The macropore structure of the top 50 mm governs the gas exchange between the soil and the atmosphere under wet conditions. In the case of nitrous oxide (N_2O), the top 50 mm of the soil was additionally shown to be the most active zone for N_2O production (Clayton *et al.*, 1994). In most soils N_2O is mainly produced as an intermediate product of denitrification, which is promoted by low O_2 concentrations that occur typically in wet soils. Nitrous oxide emissions from soils account for 70% of total N_2O emissions, which are estimated to be responsible for approximately 5% of anticipated global warming (Mosier *et al.*, 1998).

In structured soils under wet conditions, N_2O is produced in the micro-pores of soil. The O_2 concentration around the aggregates is the most important boundary condition influencing N_2O production within aggregates (Sextstone *et al.*, 1985; Tiedje, 1988). Soil microbial reactions consume O_2 , and anoxic

zones develop inside the soil structure. The volume of the anoxic zone primarily depends on the size of the interfacial area of the soil pores with O_2 (Renault & Stengel, 1994). This interfacial area is small, for example, when the pore is covered by water, thereby obstructing a rapid diffusion of O_2 into the pores. As a consequence, the anoxic volume within the pore would then be large, and would enhance N_2O production. Nitrous oxide production and emission generally rise with increasing water contents and are greatest when soils are close to water saturation (Bollmann & Conrad, 1998; Dobbie & Smith, 2001). Near saturation, only the soil's macropores are filled with air, promoting rapid exchange of O_2 . Atmospheric O_2 is exchanged over their interfacial areas with adjacent structures. Therefore, while the N_2O production happens at the scale of micro-pores within aggregates, it is regulated by the O_2 exchange processes at the scale of the soil's macropore network. Therefore, a difference in macroporosities will modify the gas exchange between the soil and the atmosphere. For example, increasing the macroporosity of wet soils results in greater rates of O_2 exchange with the atmosphere, and as a consequence the production and, therefore, the emission of N_2O may be reduced. Several field experiments have reported reduced N_2O emissions from soils containing comparably larger macroporosities (Bijay-Singh *et al.*, 1989; Teepe *et al.*, 2004). We hypothesize that the emission of N_2O from soils is an environmental risk that, *inter alia*, depends on the soil's macropore structure.

We compared the soil's macropore structures of the top 5 cm of soil in two apple orchards with different soil C management but with the same year of establishment, previous land use, soil type, texture and climate. We addressed three questions:

- 1 Are the soil's macropore structures in the two systems significantly different?
- 2 Are the biological processes that promote the formation and stabilization of macropore networks in the two systems significantly different?
- 3 Is the gas exchange around aggregates under wet conditions significantly different in the two systems?

Materials and methods

Study site

We selected a pair of soils under apple orchards in Hawke's Bay (North Island of New Zealand), one of the premier areas for horticultural production in New Zealand. The topsoils are of alluvial origin, have a silt loam texture (Table 1) and had already been studied in the context of soil quality (Vogeler *et al.*, 2006; Deurer *et al.*, 2008). One of the sites contained the tree rows of an organic orchard and the other site had the tree rows of a neighbouring conventional integrated apple production system. The organic orchard has been under organic management (BioGro™) since 1997. The apple trees in the organic orchard were 13 years old and had not been irrigated. Green-waste

Table 1 General soil characteristics of the topsoil (0–0.1 m) of the organic and integrated apple orchard systems

	Organic orchard	Integrated orchard
Texture	Silt loam	Silt loam
Sand/%	2.7	0.4
Silt/%	65.2	71.1
Clay/%	32.1	28.5
Bulk density/g cm ⁻³	1.1	1.4
pH (H ₂ O)	6.4	7.1
MWD ^a	1.82	1.33

^aMean weighted diameter of aggregates. The method of Le Bissonais (1996) was used. More details are given elsewhere (Deurer *et al.*, 2008).

compost has been applied to the topsoil of the tree rows once a year at a rate of 5–10 t ha⁻¹, and lime has been added at a rate of 300 kg ha⁻¹ every 4 years. Lime-sulphur and copper have been used as fungicides if needed. The tree rows were grassed predominantly with *Lolium perenne* L. and *Trifolium repens* L. and regularly mowed as necessary. The apple trees in the adjacent integrated orchard system were 12 years old. A 0.5-m wide strip under the trees has been kept vegetation-free by regular herbicide applications. The apple trees have been drip-irrigated during the vegetative period; 50 kg N ha⁻¹ nitrogen fertilizer as calcium ammonium nitrate has been annually applied, and lime has been added at a rate of 300 kg ha⁻¹ every 4 years. The irrigation, nutrient and pest management followed the guidelines of integrated fruit production (Wiltshire, 2003).

Analysis of the soil's 3D macropore structure

The analysis of the 3D macropore structure involved three steps: (i) sample selection and preparation, (ii) 3D X-ray computed tomography of the samples and (iii) digital image processing and analysis of the 3D macropore structures.

Sample selection and preparation. We took three undisturbed columns (70 mm diameter, 100 mm length) from the soil of each orchard system. The size of the columns was larger than the size of interest to minimize the possible interference of the excavation with the pore structure. Each of the columns was taken from one independent tree row. Within each tree row the sampling location was randomly selected. The tops of the columns were level with the soil surface and the main axis of the columns was perpendicular to the soil surface. We excavated the columns in June 2007 at a time when the water contents were at near field capacity.

After the excavation of the columns, a 1% formalin solution was infiltrated into each column to expel the earthworms from the columns to try to prevent an unnaturally intensive burrowing behaviour because of the confined space in the cores. One earthworm was expelled from each of two soil columns of the integrated orchard while the third column had no earthworms. In the organic orchard, only one earthworm was expelled from one

of the soil columns. We assumed that expelling the earthworms did not affect significantly the existing macropore structure. The soil columns were then stored at 5°C. Prior to the X-ray imaging, each soil column was carefully cut into four sub-columns of equal dimensions. Each soil column was halved by carefully cutting the column from the top to the bottom along its length-axis. The resulting two column-parts of equal dimensions were then again halved by carefully cutting each column part from the top to the bottom along its length axis in two sub-columns of identical dimensions. The size of the sub-columns was the maximum size that we could use for our 3D X-ray tomography based analysis and still achieve a spatial resolution of 86 µm. Three of the four sub-columns of each column were randomly selected for X-ray imaging.

Three-dimensional X-ray computed tomography. For X-ray tomography imaging we used a Metris X-tek Benchtop CT system (Metris X-tek Systems Ltd, Tring, UK) at the SIMBIOS Centre of the University of Abertay (Dundee, Scotland) with a 160 kV X-ray source and a 12-bit CCD camera.

In each scan 1169 angular projection images were collected and each radiograph was averaged over 32 frames. Ring artefacts were minimized during data acquisition and a 0.1 mm aluminium filter was used to reduce beam hardening. The software package CT-Pro v.1.0 (Metris X-tek Systems Ltd), which employs the filtered back-projection algorithm for CT reconstruction, was used to obtain the three-dimensional maps of attenuation coefficients from the two-dimensional angular projections. The three-dimensional images of attenuation coefficients with the isotropic voxel size of 86 µm were then translated into a continuous stack of two-dimensional 8-bit (with dynamic range of 256 greyscale values) TIFF images using the software VGStudio MAX 1.2.1 (Volume Graphics GmbH, Heidelberg, Germany). Each image slice of the stack had the thickness of 86 µm and an in-plane resolution of 86 × 86 µm.

Digital image processing and analysis. The digital image processing and analysis was conducted with the public domain software ImageJ (developed by W. Rasband at the National Institute of Health, USA; <http://rsb.info.nih.gov/ij/>), the soil-specific package of plugins for ImageJ, SCAMP, which was developed by SIMBIOS (University of Abertay, Dundee, Scotland), and a package of C-functions for image analysis QuantIM, Version 4 (Vogel, 2008). The three-dimensional macropore structures were visualized with the software OpenDX (open source software version of IBM's Visualization Data Explorer; <http://www.opendx.org/index2.php>).

The image processing and analysis consisted of four steps. First, the three-dimensional image data set of each sub-column was cropped to a length of 43 mm and a dimension of 20 × 17 mm in the plane. Each three-dimensional image data set consisted of a continuous stack of 500 two-dimensional images (TIFF slices). Each cropped two-dimensional image of the stack had a resolution of 233 × 198 pixels. By cropping the

images we avoided the analysis of artefacts caused by beam hardening and edge effects from cutting of the sub-columns. Secondly, we transformed the stack of 8-bit TIFF slices to a stack of binary images (black = pore space; white = solid phase) using a bi-level segmentation algorithm (Vogel & Kretzschmar, 1996). Next, all pores smaller than 344 μm in diameter were removed from the binary images, limiting our analysis to macropores larger than 0.34 mm. In image analysis terms, we opened (Serra, 1982) the pore structure by a spherical structuring element with a radius of 2 voxels (i.e. 172 μm). Finally, we quantified the macropore volume density (i.e. macroporosity) and the discrete macropore size distribution of the macropore networks as general descriptors of the macropore structure. The three-dimensional connectivity of macropores was quantified with the volumetric Euler-Poincaré characteristic for pores with a minimum pore diameter of 0.3 mm. The greater the value of the volumetric Euler-Poincaré characteristic, the smaller is the connectivity of the respective pore system. For convenience we use the term 'Euler number' to denote the value of the volumetric Euler-Poincaré characteristic. The theoretical background and methodological details of the respective image analysis to derive the porosity, pore-size distribution and the Euler number are given elsewhere (Vogel, 1997).

Derivation of the mean macropore radius from the macropore size distribution

From the density of macropores with specific sizes we derived the discrete cumulative distribution of fractions of macropore radii larger than 0.344 mm. The discrete distribution was then used to interpolate the mean macropore radius.

Fresh weights of anecic earthworms

The fresh weights of anecic earthworms were measured in three replicates in each of the two orchards each month from January to December 2007. For each of the replicates one out of three tree rows was chosen, and the location within the tree row was randomly selected. Prior to the measurement, vegetation was clipped and removed from the soil surface together with plant litter, and a round sampling area was demarcated with PVC rings (0.25 m²). A 1% formalin solution was applied with a watering can to the area. Depending on the infiltration rate, 5–15 litres of formalin were infiltrated over 30 minutes. Fully emerged anecic earthworms were continuously collected, transferred into sampling bottles and repeatedly rinsed with water in order to reduce mortality and then stored in partially water-filled bottles until measurement in the laboratory. After the infiltration experiment we excavated and then carefully hand-sieved a soil block (0.2 × 0.2 × 0.2 m) within each ring to identify the fresh weight of anecic earthworms not sampled. We corrected the total fresh weight of anecic earthworms by this earthworm weight on a total volume basis, and then derived the fresh weight of anecic earthworms per m² in the top 0.2 m. In the lab-

oratory, the anecic earthworms were rinsed with tap water, blotted dry on tissue paper and weighed. It was beyond the scope of our project to sample also the small endogeic earthworms that were present in the soils.

Microbial biomass

The microbial biomass of the soil was quantified monthly from January to December 2006 in soil samples (approximately 1000 cm³) from the top 0.1 m within the tree row. For each month, three samples were taken and each of the samples was taken from one independent tree row from a randomly selected site. We measured the microbial biomass according to the method of Höper (2006).

Simulation of gas diffusion through a soil's macropore network

The gas diffusion coefficient is the key property of gas exchange and depends not only on the porosity but also on the connectivity of pores (Weerts *et al.*, 2001; Tuli & Hopmans, 2004). Other studies have also used the diffusion coefficient as a meaningful and functional proxy for a soil's pore structure (Vogel *et al.*, 2002; Liu *et al.*, 2006).

We simulated the gas diffusion through segments of the three-dimensional macropore network and derived gas diffusion coefficients for the measured macropore structures. The segments were 4.3 mm long and had the dimension of 20 × 17 mm in the plane. The objective was to analyse the gas diffusion in the macropores around soil aggregates. The mean weighted diameter (MWD) of water-stable aggregates in the 0–0.1 m depth of the tree rows of the organic and integrated orchard system was 1.8 mm and 1.3 mm, respectively (Deurer *et al.*, 2008). Therefore, the 4.3 mm length of each segment was at least twice the MWD of the aggregates of the orchard soils, and was assumed to represent the relevant dimensions of gas exchange around aggregates. On the basis of the MWD, we estimated that each 'aggregate scale segment' (20 × 17 × 4.3 mm) of the organic orchard contained at least 90 aggregates and of the integrated orchard at least 195 aggregates.

The package QuantIM (Vogel, 2008) contains an algorithm for simulating gas diffusion through a continuous stack of two-dimensional binary images. It numerically solves Fick's law of diffusion in three dimensions with a finite difference approach. The algorithm extends the work of others (Crank, 1975; Horgan, 1999), and is described in detail elsewhere (Vogel *et al.*, 2002).

For the simulation, the following set of initial and boundary conditions was assumed. At the top of the segment the concentration was fixed to a constant value while the concentration at the bottom of the segment was set to zero throughout the simulation. The initial concentration within the pore space of the segment was set to zero. The vertical boundaries of the segment were assumed to be impermeable to gas. Once the gas flow at the bottom of the sample reached an asymptotic state, the relative

apparent diffusion coefficient D_r [-] was calculated by relating the simulated asymptotic diffusive flow through the macropore space at the bottom of the segment to the diffusive flow through the same segment that now consisted only of free air (=100% porosity).

Spatial variability of macropore related properties

The spatial variability of macropore related properties of macroporosity, volumetric Euler number and the relative apparent diffusion coefficient was analysed at three spatial scales. Each of the six excavated soil columns was represented by a set of three sub-columns ($43 \times 20 \times 17$ mm). We termed the set of three sub-columns with a total combined volume of $43\,860\text{ mm}^3$ as the 'column scale'. Each sub-column with a volume of $14\,620\text{ mm}^3$ represented the 'sub-column scale'. The sub-columns were divided into 10 segments ($4.3 \times 20 \times 17$ mm) and each segment with a volume of 1462 mm^3 represented the 'aggregate segment scale'.

We used the concept of the dispersion variance (Webster & Oliver, 2001) to partition the total spatial variance into three separate nested components of the soil columns, sub-columns and aggregate segments. The total spatial variance is the sum of the three separate component-specific variances. Instead of the variance we used the coefficient of variation to make it easier to compare the variability of properties with different mean values.

Statistical analysis

The software Genstat 9.1.0.150 [(PC/Windows XP) Copyright 2006, Lawes Agricultural Trust (Rothamsted Experimental Station)] and a one-way ANOVA were used to determine if the mean values of properties of the integrated and organic orchard were significantly different ($P < 0.05$). The monthly measurements of earthworm fresh weights or microbial biomass C were represented as blocks. For the characteristics of the soils' macropore structure the soil columns were represented as blocks. Each block had three replicates (sub-columns). We interpreted the differences between averages of properties to be significant if they were larger than their respective least significant differences ($P \leq 0.05$).

Results and discussion

Soil C management of the study sites

A recent study investigated the soil C management in the topsoil of the organic and integrated orchard in detail (Deurer *et al.*, 2008). In summary, the study found a significantly different soil carbon management by using the difference of the total and hot-water-extractable C as the respective measure. The long-term regular compost applications and pasture growth in the tree rows led to significantly ($P < 0.05$) larger contents of total and hot-water-extractable C in the topsoil of the organic

than in that of the integrated orchard. The hot-water-extractable C describes a microbially available 'labile' C fraction in the soil (Ghani *et al.*, 2003). The top 0.1 m of the organic orchard stored on average 3.8 kg C m^{-2} of total and 110 g C m^{-2} (monthly average) hot-water extractable C compared with that of the integrated orchard, with 2.6 kg C m^{-2} of total and 60 g C m^{-2} hot-water-extractable C.

Macropore structure

We reconstructed the detailed macropore networks in three dimensions with X-ray CT (Figure 1) and quantified the macroporosity and the macropore size distribution at the scale of the sub-columns. The average over the set of three sub-columns that represented one column yielded the estimate for the respective column scale (Table 2).

The organic orchard ($7.5 \pm 2.1\%$ by volume) had a significantly ($P < 0.05$) greater macroporosity than the integrated system ($2.4 \pm 0.5\%$ by volume) (Table 2). The density of macropores did not decrease in the orchard systems in the top 50 mm of the soil (Figure 2). Other studies with a similar definition of macropores reported similar macroporosities ranging from 1.3 to 14% for fine-textured topsoils under arable or pastoral land use (Larsson & Jarvis, 1999; Perret *et al.*, 1999; Ersahin *et al.*, 2002; Haws *et al.*, 2004). Another study (Vogel *et al.*, 2002) analysing the macropores larger than 0.3 mm of a sub-soil horizon found a greater macroporosity of approximately 17% (by volume).

The macroporosity in the tree rows of both orchards at 0–0.1 m depth was estimated in an earlier study as the total porosity minus the volumetric water content at -100 hPa (Vogeler *et al.*, 2006). They also found a significantly greater macroporosity in the organic (12%) than in the integrated (7%) apple orchard. A combination of both data sets suggests that in the organic orchard at least 70% and in the integrated orchard only 40% of the pore space that is air-filled at -100 hPa is contributed by pores larger 0.3 mm.

Our data suggest that the soil C management of the organic orchard, the long-term application of compost plus grass cover, leads to more SOC and greater macroporosity. Another study focusing on top-soils that varied in texture and organic matter under arable land use, however, came to the opposite conclusion (Jarvis *et al.*, 2007): SOC was a significant indicator for macroporosity and negatively correlated with the density of continuous macropores.

The mean macropore radius was similar in both of the presented systems (Figure 3). The organic orchard's mean macropore radius was $0.41 \pm 0.02\text{ mm}$ and that of the integrated orchard was $0.39 \pm 0.01\text{ mm}$ (Table 2). The soil C management did not affect the mean macropore size but did affect their density. The same biotic and abiotic processes such as root growth, drying-rewetting and meso- and macrofaunal activities form the macropores in both systems and, therefore, might explain why both production systems had a similar mean macropore

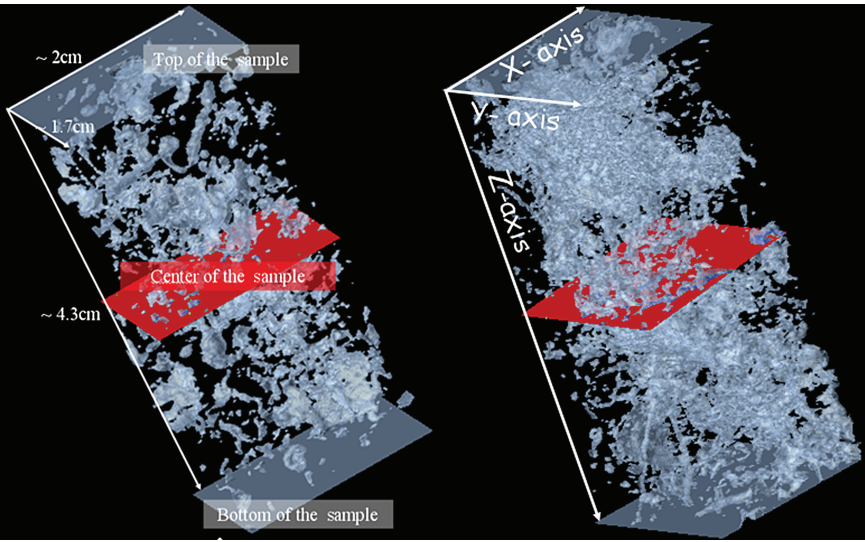


Figure 1 Examples of macropore networks in the top 50 mm of soil in the tree rows of two apple orchard systems in Hawke’s Bay, New Zealand. The grey coloured areas are macropores. The three x-y planes are shown to mark the top, centre and bottom of the sample. Left: Macropore network of the integrated orchard system. The macroporosity is 2.9% (volume basis) and the mean macropore radius 0.38 mm. Right: Macropore network of the organic orchard system. The macroporosity is 8.3% on a volume basis and the mean macropore radius 0.41 mm.

diameter. The fraction of very large pores with a diameter larger than 2 mm was also not significantly different in the two systems. Pores with a diameter larger than 2 mm contributed, at the column scale, $1.48 \pm 0.46\%$ of the total macroporosity in the organic orchard and $1.32 \pm 0.33\%$ in the integrated orchard. Our values of a mean macropore diameter of approximately 0.8 mm correspond with the range of values of conductive pore diameters that were derived from surface infiltration measurements near saturation (Sauer *et al.*, 1989). The latter study compared fine-textured soils under different arable land uses

and found a conductive pore diameter of 1.34 ± 0.67 mm for a no-till and 0.19 ± 0.18 mm for a ploughed soil treatment.

We quantified the connectivity of macropores with the volumetric Euler number at the scale of aggregate scale segments and averaged the aggregate scale values, first over the sub-columns and second over the set of three sub-columns that represented one column to estimate the column scale values (Table 2). The column scale connectivity tended to be greater in the organic compared with the integrated orchard system (Table 2). However, this was not statistically significant as the variability of the Euler number in the organic orchard was extremely large (Table 2). The range of Euler numbers for pores with a diameter larger than 0.3 mm agreed well with other comparable studies (Vogel, 1997; Vogel *et al.*, 2002).

Table 2 Biophysical properties of the integrated and organic orchard soils. The standard deviation of the mean is given in brackets

Property	Integrated orchard	Organic orchard
Macroporosity ^a /% volume	2.4 (0.5)	7.5 (2.1)
Mean macropore radius ^a /mm	0.41 (0.02)	0.39 (0.01)
Euler number ^a /mm ⁻³	0.0196 (0.0048)	0.0066 (0.029)
Relative diffusion coefficient ^a /-	0.0056 (0.0009)	0.0239 (0.0079)
Microbial biomass carbon ^{b,c} /g m ⁻²	73 (16)	143 (40)
Earthworm fresh-weight ^{b,d} /g m ⁻²	85 (53)	154 (47)

^aAt the soil column scale.
^bMonthly average of 2006.
^cIn the top 0.1 m of the soil.
^dIn the top 0.2 m of the soil.

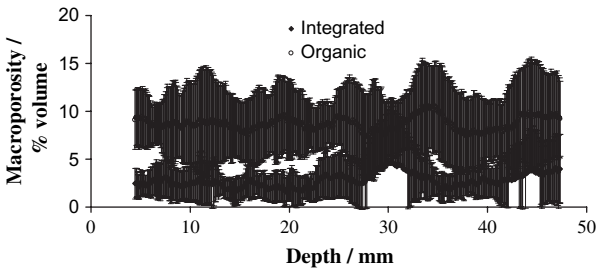


Figure 2 The macroporosity as a function of depth below the soil surface in the tree rows of an organic and integrated orchard system. The symbols show the means and the bars one standard deviation of the nine sub-columns of each system. The macroporosity was $7.5 \pm 2.1\%$ (volume) in the organic orchard and $2.4 \pm 0.5\%$ in the integrated orchard.

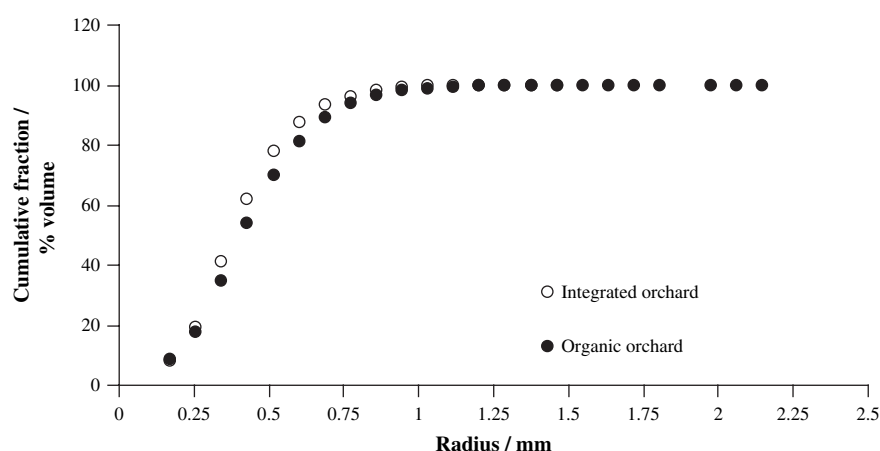


Figure 3 The macropore radius size distribution of two example sub-column slices of the integrated and organic orchard systems. The mean macropore radius is derived as the macropore radius at a cumulative fraction of 50%. In this example the mean macropore radius of the sub-column of the integrated orchard was 0.38 mm and of the organic orchard 0.41 mm.

Formation and stabilization of macropores

Soil C management and the increase of SOC can only indirectly affect a soil's macropore structure by promoting the biogenic formation and stabilization of macropores. Therefore, we conducted additional measurements to confirm this indirect impact of soil C management on the soil macropore structure. We assumed that the pasture growing in the tree rows of the organic orchard led to a greater root density in the top 50 mm of the soil. The surface vegetation growth in the tree rows of the integrated orchard was regularly controlled with herbicides.

Another important source of macropores is anecic earthworms. Anecic earthworms can produce several hundred channels per m² that are in the range of 2–12 mm in diameter (Edwards *et al.*, 1988; Pittikaenen & Nuutinen, 1997; Shipitalo & Butt, 1999). We measured the fresh weight of anecic earthworms each month as an indicator of their burrowing activity over 1 year (Figure 4). On average, the fresh weight was significantly ($P < 0.05$) greater in the organic orchard (154 ± 47 g m⁻²) than in the integrated orchard (85 ± 53 g m⁻²) (Figure 4). Therefore, the soil C management led to greater densities of anecic earthworms in the organic orchard. Generally, biopores generated by earthworms can be highly dynamic, with a short turn-over time (Daniel *et al.*, 1997).

The macroporosity is not only a function of the formation but also of the stabilization of macropores. The stability of macropore networks depends, *inter alia*, on the stability of aggregates. In soils with similar texture, the stability of macroaggregates usually increases with C content and microbial activity (Le Bissonais & Arrouays, 1997). There was a significantly greater aggregate stability and microbial activity (Figure 4) in the organic than in the integrated orchard (Deurer *et al.*, 2008). The aggregate stability as indicated by MWD (Table 1) was 1.82 mm in the organic and 1.33 mm in the integrated orchard (Deurer *et al.*, 2008). The monthly average of

microbial biomass C was 143 ± 40 g microbial biomass C m⁻² in the organic orchard, which was significantly larger than the monthly average in the integrated system (73 ± 16 g C m⁻²) (Deurer *et al.*, 2008). Overall, the pronounced soil C management in the organic orchard increased both the formation of macropores by roots and anecic earthworms and the stabilization of the macropore structure by a larger aggregate stability and microbial biomass.

Relative diffusion coefficients

From the gas exchange simulations around the aggregates in the two topsoils for wet conditions significantly larger relative diffusion coefficients were derived for the organic, than for the integrated system. The relative diffusion coefficient at the column scale was 0.0239 ± 0.0079 in the organic and 0.0056 ± 0.0009 in the integrated system, respectively (Figure 5). We suggest that the larger relative diffusion coefficients in the soil of the organic orchard indicate less favourable physical conditions for N₂O production and emission. Therefore, this indicates that soil C management possibly combats climate change in two ways. It firstly leads to C sequestration, and secondly might also contribute indirectly to a reduction in N₂O emissions. Research that measures N₂O emissions from topsoils that differ ideally only with respect to their macropore structure is needed to prove or disapprove this hypothesis.

Relative diffusion coefficients are scale-dependent (Vogel *et al.*, 2002). In our study each segment mimicking the aggregate scale had a volume of 1462 mm³. The relative diffusion coefficient of the organic orchard was at this scale very similar to the value of 0.02 found by others (Vogel *et al.*, 2002) at the same volume scale using the same definition of macropore and the same method to simulate diffusion. Vogel *et al.* (2002) worked with a loamy-clay sub-soil under arable land use. However, the macroporosity in their study was, by about

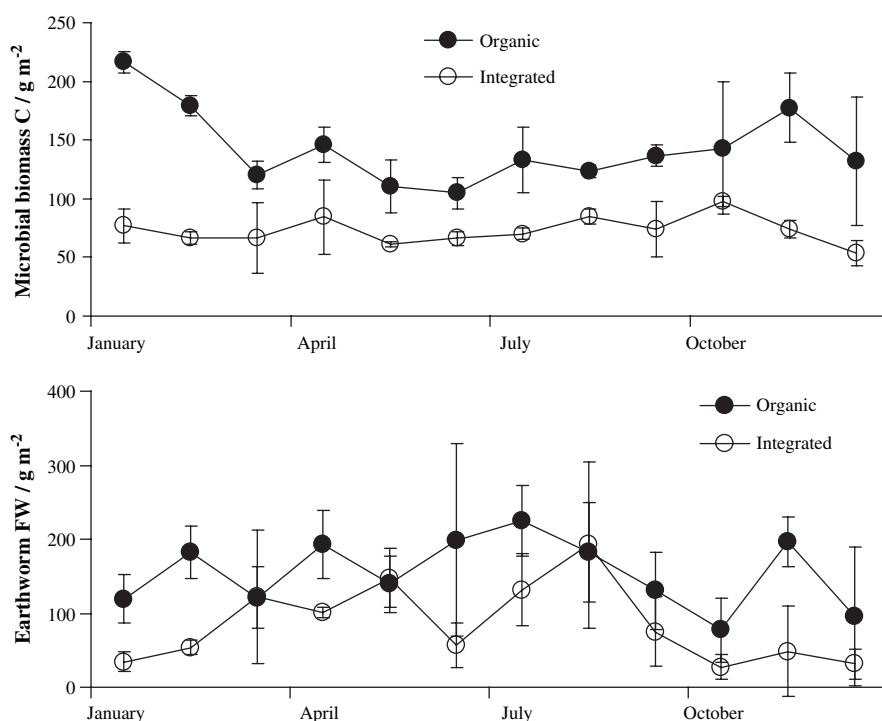


Figure 4 Soil biological indicators for macropore stabilization (microbial biomass C) and formation (fresh weight of earthworms). Top: Monthly microbial biomass C in the top 0.1 m in 2006. The bars denote one standard deviation. These measurements were already carried out by Deurer *et al.* (2008), who only published their yearly average. Bottom: Monthly fresh weights of anecic earthworms in the top 0.2 m in 2006. The bars denote one standard deviation.

a factor of two, greater at 17% on a volume basis. The fact that the magnitude of the relative diffusion coefficients of the two studies is still comparable suggests that the connectivity and continuity of macropores is very important, and that the top-soil of our system had a better connectivity and continuity of macropores than the sub-soil used by Vogel *et al.* (2002). Other studies have also stressed that the extent of the connectivity of the gas-filled pore space controls the transport in the

gaseous phase, especially at the wetter end of the soil water content range (Weerts *et al.*, 2001; Tuli & Hopmans, 2004). In a multiple regression, 76% of the variability of the relative diffusion coefficient was explained in the integrated orchard with macroporosity and the Euler number as the significant variables (Table 3). However, in the regression for the organic orchard only the macroporosity was a significant variable and explained 71% of the variability (Table 3).

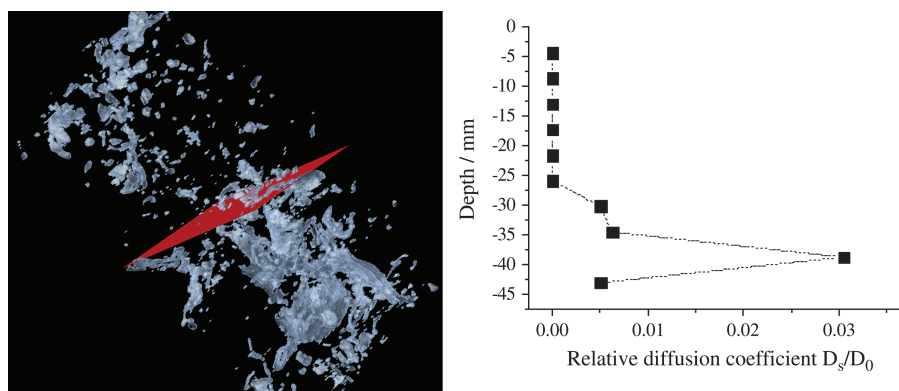


Figure 5 Left: The macropore structure of an example sub-column of the integrated apple orchard soil. The sub-column is 43 mm long and the in-plane dimensions are 20 × 17 mm. Right: The respective aggregate-scale relative diffusion coefficients of the sub-column as a function of the depth below the soil surface.

Table 3 Results of multiple linear regressions of the relative diffusion coefficients D_R [-] (dependent variable) at the aggregate segment scale in the soils of the integrated and organic orchard soils. The macroporosity and the Euler number were used as the independent variables

	Integrated orchard	Organic orchard
Y-intercept/-	-0.0017 ^{n.s.}	-0.0172*
Macroporosity/% volume	0.5024*	0.551*
Euler number/mm ⁻³	-0.2504*	— ^a
R^2	0.76	0.71
Root mean square error	0.0069	0.0121

n.s., not significant.

*Highly significant ($P < 0.001$).

^aThe Euler number was not a significant factor. We used a simple linear regression instead of a multiple linear regression.

The relative diffusion coefficient of the integrated system was small. Indeed, 38 of the 90 aggregate-scale segments or 42% of all segments had a relative diffusion coefficient of zero because of a lack of macropore continuity. Also, all nine sub-columns of the integrated orchard had at least one segment with a relative diffusion coefficient of zero (Figure 5). Therefore, the diffusion upwards from a depth of approximately 50 mm would have been zero for all nine sub-columns of the integrated orchard and, under wet conditions, the oxygen exchange with the atmosphere would be limited (Figure 5).

Spatial variability of macropore related properties

Macropore related properties are scale-dependent and it is probably not possible to define one unique representative elementary volume for them (Vogel *et al.*, 2002). For example, with the small sample size used we could not capture representatively large macropores of the orchard soils such as channels generated by the decay of tree roots. We assessed the impact of soil C management on the soil's macropore network at specific spatial scales and used the aggregate scale diffusion as an example of soil function that is meaningful at these scales.

We estimated how much the column, sub-column and aggregate scale contributed to the overall spatial variability of macropore related properties (Table 4). With the exception of the relative diffusion coefficient, different scales dominated the vari-

ability of macropore related properties in the two orchards. The macroporosity varied mainly at the aggregate scale in the integrated orchard and at the column scale in the organic orchard. The relative diffusion coefficient was dominated by the heterogeneity of the aggregate scale in both orchard systems (Table 4). The Euler number showed a similar variation at the three scales in the integrated orchard and varied mainly at the column scale in the organic orchard (Table 4). Therefore, at the spatial scales examined, the orchard-specific soil carbon management generated not only quantitatively different macropore related properties but had also an impact on their spatial variation.

The coefficient of variation of the relative diffusion coefficients at the aggregate scale was 70% in the organic orchard and 188% in the integrated orchard, respectively (Table 4). We suggest that the extremely large variability of diffusion observed in the integrated orchard system could explain the commonly observed extremely high spatial and temporal variability of N_2O emissions from soils. For example, the coefficient of variation of N_2O emissions from soils typically ranges from 70 to 370% (Parkin, 1987; Mathieu *et al.*, 2006). Various biochemical factors such as SOC and nitrate concentrations influence N_2O production. However, the variation of these factors is much smaller than that of N_2O emission from soils (Robertson & Groffman, 2007). Also, attempts to explain the variability of N_2O emissions as a function of bulk soil properties have only been moderately successful (Flessa *et al.*, 1995; Velthof & Oenema, 1995; Mathieu *et al.*, 2006). However, the macropore structure has, to our knowledge, never been used as an explanatory variable for N_2O emissions.

Conclusions

We studied how the long-term OC addition (compost plus grass cover) and the related increase of SOC contents affected the soil macropore structure and the near-surface gas exchange under wet conditions. The tree rows of an organic orchard with soil C management were compared with those of an integrated orchard without soil C management. The long-term additions of OC changed the macropore structure: the organic orchard had significantly greater macroporosities. The mean macropore radius was similar in both systems. The connectivity of macropores tended to be greater in the organic orchard but this was not statistically

Table 4 Partitioning of the total spatial variation of key macropore properties of the integrated and organic orchard soils into its scale-dependent components. The considered spatial scales are the column (CS, $N = 3$, 43 860 mm³), the sub-column (SCS, $N = 9$, 14620 mm³), and the aggregate segment scale (ASS, $N = 90$, 1462 mm³). The spatial variation was quantified with the coefficient of variation. The numbers in brackets give the relative fraction that the variation of each scale contributes to the total variation (= value of 1)

Coefficient of variation/%	Integrated orchard			Organic orchard		
	CS	SCS	ASS	CS	SCS	ASS
Property						
Macroporosity/% volume	21 (0.21)	21 (0.21)	58 (0.58)	28 (0.7)	5 (0.13)	7 (0.17)
Euler number/mm ⁻³	24 (0.38)	19 (0.30)	20 (0.32)	440 (0.63)	33 (0.05)	229 (0.32)
Relative diffusion coefficient/-	16 (0.06)	61 (0.23)	188 (0.71)	33 (0.26)	23 (0.18)	70 (0.56)

significant. It is assumed that similar biotic and abiotic processes such as root growth, drying-rewetting and meso- and macrofaunal activities formed the macropores in both systems. We showed that the formation of macropores, as indicated by roots and anecic earthworms, and the stabilization of macropores, as indicated by the aggregate stability and microbial biomass, were significantly greater in the organic orchard than in the integrated orchard. The indicators of the formation and stabilization of macropores in turn are known to depend on the soil C management.

In most soils N_2O is produced mainly as an intermediate product of denitrification. Denitrification is promoted by low O_2 concentrations that typically occur in wet soils. We simulated the gas exchange through the measured three-dimensional macropore network around aggregates and found significantly larger relative diffusion coefficients in the organic than in the integrated orchard. In the integrated orchard the spatial variability of relative diffusion coefficients was mainly a function of the porosity and connectivity and in the organic orchard a function of the porosity of macropores. Those macropore properties in turn were influenced by the soil C management. We suggest that larger relative diffusion coefficients under wet conditions indicate less favourable physical conditions for N_2O production and emission. Therefore, soil C management possibly combats climate change in two ways. It firstly leads to C sequestration and secondly might also contribute indirectly to a reduction of N_2O emissions, by creating more and better connected macropores near the soil surface.

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